

Healthcare Business Capstone

Hospital Operations & Revenue Risk Intelligence Platform

This capstone project is designed as an end-to-end, industry-aligned learning journey that simulates how modern analytics and AI systems are designed, built, evaluated, and deployed in real-world healthcare organizations. Learners will work on a realistic, business-focused hospital scenario involving a multi-specialty hospital network operating across multiple cities.

The capstone emphasises the application of SQL, Python, Machine Learning, and MLOps practices to solve operational and financial challenges faced by hospital leadership, clinical teams, and finance departments.

Capstone Overview

Large hospital networks face increasing challenges related to patient flow management, operational inefficiencies, and revenue leakage caused by insurance claim rejections and delayed payments. These challenges arise due to fragmented data systems, limited visibility into operational risks, and the absence of predictive intelligence in decision-making processes.

The Healthcare Business Capstone addresses these challenges through a structured, multi-phase project that mirrors how analytics and AI teams build production-grade decision-support systems in healthcare environments. The focus is on creating solutions that are actionable, interpretable, and aligned with hospital business objectives.

Capstone Structure

The capstone is divided into multiple connected phases. Each phase focuses on a different aspect of hospital operations, financial performance, and AI system maturity, while collectively building a unified hospital risk and revenue intelligence platform.

Phase 1: SQL Analytics Layer (Business Intelligence Foundation)

In Phase 1, learners focus on transforming raw hospital operational and financial data into a reliable, structured SQL analytics layer. This phase emphasizes relational database design, data integrity enforcement, indexing strategies, and reusable business queries.

Learners create SQL views and analyses that enable hospital leadership to monitor department performance, patient flow efficiency, insurance provider behavior, and revenue realization. This phase establishes a trusted data foundation for all downstream analytics and modeling efforts.

Phase 2: Exploratory Data Analysis & Data Quality (Analytics Intelligence)

Phase 2 transitions the workflow into Python-based exploratory data analysis. Learners combine patient, visit, and billing datasets to uncover trends, risks, inefficiencies, and data quality issues.

This phase introduces missing value analysis, distribution analysis, outlier detection, and business-driven feature engineering. The focus is on understanding hospital operations and

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financial dynamics before applying predictive models, ensuring that insights are reliable and decision ready.

Phase 3: Model Development (Classification Systems)

In Phase 3, learners build machine learning classification systems to support proactive hospital decision-making.

Model A: Visit Risk Classification

Learners develop a model to classify hospital visits into Low, Medium, or High risk categories based on operational and clinical indicators. This model supports resource planning, patient prioritization, and operational risk management.

Model B: Claim Outcome Classification

Learners build a predictive model to estimate insurance claim outcomes (Paid, Pending, or Rejected) before submission. This model helps finance teams reduce revenue leakage and improve cash flow predictability.

Both models emphasise time-based validation, business-relevant feature selection, and responsible evaluation practices.

Phase 4: Model Evaluation & Explainability

Phase 4 focuses on ensuring that model predictions are interpretable, reliable, and safe for healthcare use cases. Learners evaluate models using both technical metrics (precision, recall, F1-score) and business-critical metrics such as recall for High Risk visits and Rejected claims.

This phase introduces explainability techniques, fairness analysis across demographics and insurance providers, and consolidated model documentation through a model card approach.

Phase 5: Model Deployment & API Integration (MLOps)

In Phase 5, learners operationalize trained models by deploying them as API services. The objective is not further performance optimization, but production readiness.

Learners build FastAPI-based services, implement schema validation and log predictions with versioning metadata. This phase simulates how hospital analytics solutions are integrated into dashboards and internal systems.

Phase 6: Monitoring, Drift Detection & Governance

Phase 6 focuses on long-term system reliability, governance, and compliance. Learners implement data validation checks, monitor feature and prediction drift, and maintain audit logs for regulatory and operational transparency.

This phase reinforces best practices for responsible AI deployment in healthcare environments, including documentation of assumptions, limitations, and retraining strategies.

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Final Phase: Executive Business Presentation

The final phase requires learners to translate technical outcomes into leadership-level business insights. The emphasis is on storytelling, ROI justification, and strategic alignment.

Learners present:

- Hospital business problems and operational risks
- End-to-end system architecture and data flow
- Key insights from SQL and EDA
- Model performance in business terms
- Financial impact and revenue optimization potential
- Deployment, scaling, and risk management strategy

Key Learning Outcomes

By completing this capstone, learners will be able to:

- Design end-to-end analytics and AI solutions for healthcare operations
- Build and evaluate classification models with business impact in mind
- Detect and mitigate data quality and leakage risks
- Translate predictive outputs into operational and financial decisions
- Deploy, monitor, and govern AI systems using modern MLOps practices
- Understand how analytics, machine learning, and deployment coexist within a single hospital intelligence platform

How to Approach This Capstone

Learners should approach this capstone as a simulation of professional healthcare analytics work rather than a collection of disconnected assignments. Each phase of the project must be clearly documented, technically justified, and interpreted in both business and clinical contexts to demonstrate real-world applicability.

The final deliverable should be a cohesive Hospital Operations & Revenue Risk Intelligence Platform that integrates analytics, machine learning, and deployment workflows and can be realistically proposed to hospital leadership as a deployable solution.